

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 1.1 TAXONOMY I: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the whole unit. Our driving question was: How can we organise the many different living organisms around us so that everyone can identify and study them in the same way? Think back to Lesson 1, when your group sorted a goat, a chicken, a maize plant, a mushroom, algae, bacteria, a bean plant and a mosquito, and found that other groups sorted them differently and used different names in Kikuyu, Luo, Kiswahili and English. Answer each section in the space under the prompt, using full sentences. Use evidence from the activities you did (sorting, case sets, rank cards, name comparison cards, kingdom profile cards). Use the Claim-Evidence-Reasoning (CER) structure where the prompt asks for it. Write scientific names correctly. Your group poster or classification guide of local organisms and your own written explanation will be marked on: correct use of characteristics, evidence and justification, accuracy of classification and naming, communication and collaboration, and local application and conservation.

Part 1: Why our first groupings were not enough

Look back at the Lesson 1 sorting activity with the eight organisms. (a) Describe two different ways groups sorted the organisms and what you noticed when groups compared. (b) Why was every group correct by its own rules, yet the groupings could not be used by everyone in the same way? (c) Explain why local names alone are not enough for scientists, using what you found about names for the same organism in Lesson 1.

(a) One group sorted by habitat (where the organisms live) and another sorted by usefulness (food, pests, disease), and others used colour, size, or whether the organism is green or can move. When we compared, the same organism, for example the mosquito or the maize plant, was in different groups, and the same organism had different names in Kikuyu, Luo, Kiswahili and English. (b) Each group was correct by its own criteria, but the criteria and names were different, so people could not be sure they were talking about the same organism or grouping organisms in the same way. There was no shared system. (c) In Lesson 1 we found that the same organism has different local names in Kikuyu, Luo, Kiswahili and English. Because names differ between communities, people cannot be sure they are discussing the same organism, so local names cannot be relied on by everyone. Scientists need a shared system to identify, study and communicate about organisms correctly, and correct identification also helps in conserving biodiversity.

Part 2: The hierarchy of taxonomic units

(a) List the seven taxonomic ranks in order from the broadest to the most specific. (b) Choose one Kenyan

(a) Kingdom, Phylum (Division in plants), Class, Order, Family, Genus, Species. (b) The ranks are nested, each one sitting inside the one above it. Take the maize plant (*Zea mays*). At Kingdom level it sits in a very large group with all plants. At each step down, through Phylum/Division, Class, Order, Family and Genus, the group it belongs to contains fewer organisms, and the organisms in the group share more features, so they are more similar to each other. At Species level the group is the smallest and most specific,

organism such as the goat or the maize plant (<i>Zea mays</i>) and follow it from Kingdom down to Species. Explain what happens to the number of organisms in a group and how similar they are as you move down. (c) Write a CER statement: Claim, Evidence and Reasoning for why a ranked hierarchy helps people in Nairobi, Kisumu and Mombasa identify an organism in the same way.	containing only maize plants of the same kind. (c) Claim: A ranked hierarchy lets everyone place an organism in the same position. Evidence: When we arranged the rank cards and classified the goat, chicken, maize, bean plant, mosquito and mushroom, some shared the same Kingdom or Phylum but separated at lower levels, and the placement did not depend on the organism's name in Kikuyu, Luo, Kiswahili or English. Reasoning: Because the groups are based on shared characteristics and arranged in the same order everywhere, anyone using the same chart gets the same placement, unlike sorting by colour, habitat or usefulness, which gives different groups.
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Part 3: One name for every organism

(a) Explain why common names cause problems when people share information, using maize, beans, goat or mosquito as an example. (b) State the rules for writing a scientific name using binomial nomenclature. (c) Here are three names written wrongly: 'zea Mays', 'capra hircus' and 'Anopheles Gambiae'. Rewrite each correctly and say which rule was broken in each name.	(a) Common names vary with language and community; for example, maize, goat and mosquito each have different names in Kikuyu, Luo, Kiswahili and English. This causes confusion, because people may not be sure they mean the same organism. A universal name avoids this. (b) A scientific name has two parts: the genus and the species. The genus begins with a capital letter and the species begins with a small letter. The name is in Latin form, so it is the same worldwide, and it is written in italics, or underlined when handwritten, with the genus and species underlined separately. Every organism has one agreed scientific name. (c) 'zea Mays' should be <i>Zea mays</i>: the genus 'zea' should start with a capital letter, and the species 'Mays' should start with a small letter, so both parts broke the capitalisation rule. 'capra hircus' should be <i>Capra hircus</i>: only the genus was wrong, because it should start with a capital letter. 'Anopheles Gambiae' should be <i>Anopheles gambiae</i>: only the species was wrong, because it should start with a small letter. Each correct name should also be written in italics, or if handwritten, with the genus and the species underlined separately.
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Part 4: Placing organisms into the five kingdoms

Choose any four of these organisms from the final challenge: <i>Spirogyra</i>, <i>Escherichia coli</i>, <i>Rhizopus</i> on stale bread, <i>Agaricus</i> (mushroom), <i>Zea mays</i>, tilapia (<i>Oreochromis niloticus</i>), <i>Anopheles</i> mosquito, <i>Amoeba</i>, <i>Acacia</i>. For each one, name its kingdom out of the five kingdoms, and give the observable evidence (such as	Any four correct choices are acceptable. Examples: <i>Escherichia coli</i> is in Monera (bacteria): it has a prokaryotic cell with no true nucleus and is single-celled. <i>Spirogyra</i> is in Protocista: its cells have a true nucleus and chlorophyll, and it makes food by photosynthesis. <i>Amoeba</i> is in Protocista: single-celled with a true nucleus, no cell wall and no chlorophyll, and it feeds by taking in food. <i>Rhizopus</i> on stale bread and <i>Agaricus</i> (mushroom) are in Fungi: they have cell walls, have no chlorophyll and absorb food from dead or decaying material. <i>Zea mays</i> and <i>Acacia</i> are in Plantae: many cells, cell walls, chlorophyll, and they make their own food by photosynthesis. Tilapia and the <i>Anopheles</i> mosquito are in Animalia: many cells, no cell walls, no chlorophyll, and they feed on other organisms. Characteristics such as cell type, number of cells, cell wall, chlorophyll and mode of nutrition are shared features that anyone can observe and check, so everyone gets the same answer. Colour, habitat or usefulness differ between groups and people, so they give different groupings.
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cell type, number of cells, cell wall, chlorophyll and mode of nutrition) that supports your decision. Then explain why these characteristics are better for classifying than colour, habitat or usefulness.	
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Part 5: Our final explanation and why taxonomy matters

Write your Final Explanation answering the driving question: How can living organisms be systematically organised, named and identified? Use a CER structure. Include: how organisms are grouped, how they are named, how you can identify the kingdom of an organism, and why this matters for people in Kenya (for example disease vectors, crop pests, useful organisms, and conserving biodiversity). Finally, say how your thinking has changed from your Lesson 1 model.	Claim: Living organisms can be organised, named and identified in the same way everywhere by using classification based on shared characteristics, a hierarchy of ranks, and binomial nomenclature. Evidence: In Lesson 1 our groups sorted the goat, chicken, maize plant, mushroom, algae, bacteria, bean plant and mosquito by habitat, colour, size or usefulness and got different groups, and the organisms had different names in Kikuyu, Luo, Kiswahili and English. Later we arranged organisms into the seven ranks from Kingdom to Species, wrote scientific names such as Zea mays, Capra hircus and Anopheles gambiae, and placed unfamiliar organisms into the five kingdoms (Monera, Protocista, Fungi, Plantae, Animalia) using cell type, number of cells, cell wall, chlorophyll and mode of nutrition. Reasoning: There is a very large diversity of organisms, so we need an organised system to study them. Taxonomy identifies, names and groups organisms by their similarities. Each level of the hierarchy is more specific, and each species has one two-part scientific name (genus capitalised, species in small letters) understood worldwide. Because everyone uses the same evidence and names, people in Nairobi, Kisumu and Mombasa and across the world can identify, study and communicate about the same organism. This matters in Kenya for recognising disease vectors such as the Anopheles mosquito, crop pests, and useful organisms, for understanding the economic importance of Monera, Protocista and Fungi, and for conserving biodiversity. My thinking has changed: in Lesson 1 my model only had groups made by my own rules, but my final model has ranked levels, the five kingdoms and universal scientific names.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Correct use of characteristics (30%)	Consistently uses observable characteristics (cell type, number of cells, cell wall, chlorophyll, mode of nutrition) correctly to place organisms in kingdoms and explains why they are better than colour, habitat or usefulness.	Uses most characteristics correctly with minor errors or incomplete explanation of why they are better than personal criteria.	Uses few or incorrect characteristics, or relies on colour, habitat or usefulness.
Evidence and justification (25%)	Claim-Evidence-Reasoning is complete, with specific evidence from the lessons (sorting, case sets, rank cards, name cards, profile cards) linked clearly to the claim.	CER is present with some evidence from the lessons, but the reasoning is partly missing or general.	Claim is given with little or no evidence or reasoning.

Accuracy of classification and naming (20%)	Ranks are in the correct order, kingdoms are correct, and scientific names follow all rules (capital genus, lower-case species, italics or underlined, two parts).	Mostly accurate with one or two errors in ranks, kingdoms or name writing.	Several errors in ranks, kingdoms or the writing of scientific names.
Communication and collaboration (15%)	Writing and group poster are clear, logical and well organised, using correct vocabulary, with clear evidence of contributing to the group.	Writing is mostly clear with some correct vocabulary; contribution to group work is shown.	Writing is unclear or disorganised, with little correct vocabulary or little evidence of group contribution.
Local application and conservation connection (10%)	Uses Kenyan organisms and places and clearly explains how taxonomy helps with disease vectors, crop pests, useful organisms and conserving biodiversity.	Gives a Kenyan example and a general link to the importance of taxonomy or conservation.	Little or no link to Kenyan examples or the importance of taxonomy.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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